STAT 2020 Discussion Slides

Week 11: Delta Method and CLT

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Agenda

- Propagation of error (Delta Method)
- Two approximation cases:
  - μ_X known
  - μ_X unknown (plug-in)
- Engineering notation refresher
- CLT decision rules and examples
- Optional quick review of key distributions
- Optional differentiation/integration drills

Delta Method: Core Setup

Let X be a random variable with mean μ_X and variance σ_X^2. For a smooth function $g(\cdot)$, define

$$Y = g(X).$$

First-order Taylor expansion around μ_X:

$$g(X) \approx g(\mu_X) + g'(\mu_X)(X - \mu_X).$$

Therefore,

$$\mathbb{E}(Y) \approx g(\mu_X), \qquad \mathbb{V}(Y) \approx \big(g'(\mu_X)\big)^2 \sigma_X^2.$$

Case 1: μ_X Known

If μ_X is known, evaluate derivative at the true mean:

$$\mathbb{V}(g(X)) \approx \left[g'(\mu_X)\right]^2 \mathbb{V}(X).$$

If $X = \bar{X}$ from sample size n, then

$$\mathbb{V}(\bar{X}) = \frac{\sigma^2}{n} \quad \Longrightarrow \quad \mathbb{V}(g(\bar{X})) \approx \left[g'(\mu)\right]^2 \frac{\sigma^2}{n}.$$

Standard error form:

$$\mathrm{SE}\big(g(\bar{X})\big) \approx \left|g'(\mu)\right| \frac{\sigma}{\sqrt{n}}.$$

Case 2: μ_X Unknown (Plug-in)

In practice, μ_X is often unknown. Use plug-in estimate $\bar{X}$ for μ_X:

$$g'(\mu_X) \approx g'(\bar{X}).$$

Then estimate

$$\widehat{\mathbb{V}}\big(g(\bar{X})\big) \approx \left[g'(\bar{X})\right]^2 \frac{s^2}{n},$$

and

$$\widehat{\mathrm{SE}}\big(g(\bar{X})\big) \approx \left|g'(\bar{X})\right| \frac{s}{\sqrt{n}}.$$

Engineering Notation (Quick Refresher)

For measured quantity x with uncertainty Δx:

$$x = x_0 \pm \Delta x.$$

Relative (fractional) error:

$$\frac{\Delta x}{x_0}.$$

If $y = g(x)$, first-order propagation gives

$$\Delta y \approx \left|g'(x_0)\right| \Delta x, \qquad \frac{\Delta y}{|y_0|} \approx \left|\frac{x_0 g'(x_0)}{g(x_0)}\right| \frac{\Delta x}{|x_0|}.$$

Example: $g(x) = \sqrt{x}$

Suppose $\bar{X}$ estimates $\mu = 100$, with $\mathbb{V}(\bar{X}) = 4$. Let $Y = g(\bar{X}) = \sqrt{\bar{X}}$.

$$g'(x) = \frac{1}{2\sqrt{x}}, \qquad g'(100) = \frac{1}{20}.$$

Approximate variance:

$$\mathbb{V}(Y) \approx \left(\frac{1}{20}\right)^2 \cdot 4 = \frac{1}{100} = 0.01.$$

So

$$\mathrm{SE}(Y) \approx 0.1.$$

CLT: Need It or Not?

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ, variance σ^2.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

CLT says Z is approximately $\mathrm{N}(0,1)$ for large n.

Decision rule for discussion:

- If population is normal: $\bar{X}$ is exactly normal for any n.
- If population is not normal and $n < 30$: do not force CLT approximation.
- If population is not normal and $n \geq 30$: CLT approximation is usually reasonable.

Example A: X Normal, Small n

If $X \sim \mathrm{N}(\mu, \sigma^2)$, then

$$\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

for every sample size n, including $n < 30$.

Key point:

- This result is exact.
- CLT is not needed to justify normality of $\bar{X}$ here.

Example B: X Not Normal

Suppose X is right-skewed.

- If $n = 12$: distribution of $\bar{X}$ may still be skewed.
- If $n = 80$: distribution of $\bar{X}$ is often close to normal by CLT.

Practical note:

- For small n with non-normal X, exact distribution of $\bar{X}$ can be hard to derive.
- In that case, avoid over-claiming normal approximation.

Quick Practice: CLT Decision

Decide if normal approximation for $\bar{X}$ is justified:

1. X normal, $n = 8$
2. X heavily skewed, $n = 18$
3. X moderately skewed, $n = 64$

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Answers:

- (1) Yes, exact normality (no CLT needed).
- (2) No, n too small for a routine CLT claim.
- (3) Yes, CLT approximation is typically acceptable.

Discrete Review: PMF, Mean, Variance

Hypergeometric $X \sim \text{Hypergeo}(N, K, n)$

$$\mathbb{E}(X) = n\frac{K}{N}, \qquad \mathbb{V}(X) = n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1}.$$

Negative Binomial $X \sim \text{NegBin}(r, p)$ (failures before r-th success)

$$\mathbb{E}(X) = \frac{r(1-p)}{p}, \qquad \mathbb{V}(X) = \frac{r(1-p)}{p^2}.$$

Poisson $X \sim \text{Pois}(\lambda)$

$$\mathbb{E}(X) = \lambda, \qquad \mathbb{V}(X) = \lambda.$$

Continuous Review: PDF, Mean, Variance

Uniform $X \sim \text{Unif}(a, b)$

$$f(x) = \frac{1}{b-a},\ a < x < b, \qquad \mathbb{E}(X) = \frac{a+b}{2}, \quad \mathbb{V}(X) = \frac{(b-a)^2}{12}.$$

Exponential $X \sim \text{Exp}(\lambda)$

$$f(x) = \lambda e^{-\lambda x},\ x > 0, \qquad \mathbb{E}(X) = \frac{1}{\lambda}, \quad \mathbb{V}(X) = \frac{1}{\lambda^2}.$$

Calculus Drill: Mean and Variance

Let $f(x) = 2x$, $0 < x < 1$.

- Verify f is a valid PDF.
- Compute $\mathbb{E}(X)$.
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Answer sketch:

$$\int_0^1 2x\,dx = 1, \quad \mathbb{E}(X) = \int_0^1 x(2x)\,dx = \frac{2}{3},$$

$$\mathbb{E}(X^2) = \int_0^1 x^2(2x)\,dx = \frac{1}{2}, \quad \mathbb{V}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

Take-Home Messages

- Delta Method gives first-order approximations for transformed means and variances.
- Distinguish clearly between μ_X known vs unknown plug-in cases.
- CLT usage depends on population shape and sample size.
- Keep distribution formulas and calculus tools ready for quick review.

In-Class Attendance Check (Quick)

Let $\bar{X}$ have mean $\mu = 25$ and variance $9/n$, with $n = 36$. Define $Y = \log(\bar{X})$. Use first-order Delta Method to approximate $\mathrm{SE}(Y)$.

Please submit: Full Name; Section Number; Your Answer

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Answer:

$$g'(x) = \frac{1}{x}, \quad \mathrm{SE}(\bar{X}) = \frac{3}{6} = 0.5, \quad \mathrm{SE}(Y) \approx \left|\frac{1}{25}\right|(0.5) = 0.02.$$